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Chapter 15

Geriatric Rehabilitation

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Abstract

Geriatric rehabilitation aims to improve or restore the motor function, sensorial and cognitive status of elderly having a loss or disability due to a disease, which may be an acute or chronic condition. However, elderly people living independently in the community or in the institutions may gain the benefits of geriatric rehabilitation in the concept of preventive measures and social rehabilitation. Thus, we may briefly define the geriatric rehabilitation as an approach of multidisciplinary interventions to upgrade the functional level and the overall quality of life as well as the life satisfaction of elderly.

In this concept, we aimed to present physical therapy and rehabilitation approaches for elderly having neuromusculoskeletal problems; the importance of the assessment, evaluation and the treatment planning from the physiotherapist point of view. Exercise regimes and fall preventions with balance, flexibility and strengthening exercises; physical therapy approaches for the patients with cognitive deficiencies and how to embed these approaches into the regular physical therapy programs are the subjects discussed. Additionally, some information about the assistive devices and assistive technology in improving the physical and mental status of the elderly, in providing a safe, comfortable and functional gait pattern and activities of daily living were also conveyed.

Introduction

As the developments in civilizations are increasing, the improvements in science and technology give rise to lengthen the life span; in this context, nowadays, the health prevention and quality of life issues are started to be considered among the basic needs of elderly population. Thus, physical, sensorial, psychological and cognitive changes that are eventually happening as aging occurs are needed to be delayed or supported by means of rehabilitative interventions. We may convey that rehabilitation of elderly is a complex process of

multidisciplinary and interdisciplinary approaches, simply to improve the function and the quality of life of the elderly in any condition. However, the rehabilitation processes are not only to give services to the elderly but also to support the family members and the caregivers, and to increase their knowledge regarding to the issues of elderly home care. The involvement of elderly, their family and the caregivers to the process of rehabilitation is essential and profitable to achieve the most possible wellness of the elderly physically, cognitively, psychologically and socially as suggested by [1].

The optimal functional level in home, in an institution or in the community is the fundamental aim of the geriatric rehabilitation. For instance, the aim may be simply to relieve the pain in knee joints due to osteoarthritis via physiotherapeutic approaches; however, the activity level of the elderly in home or in the institution is also important.

Thus, while searching pain, it is important to question whether, the elderly is spending her/his time sitting and preferring mostly staying indoors or being eager to have an active life style. In other words, the activity level and the motivations of the elderly person should be searched and her/his expectations should be well understood. Besides the personal circumstances, the environment of the elderly is important in order to improve the functional level of the elderly and to urge them to have a much more active life style. For instance, a house or a nursing home with well lightened corridors, nonslippery bathrooms, wide passages, ramps, handrails by the stairs and so on, may ease the life of the elderly in daily living but they are also important for a safe environment. Furthermore, outdoor environment should also be considered for the elderly to interact with society and continue to have an active life style as a part of the society with vocational, recreational and leisure activities. Hence, the aim of the geriatric rehabilitation is to maximize of the functional level, mobility, and independence of elderly, in order to increase their quality of life (QoL) [2, 3] not only from a medical point of view but also from the social aspect [4-6].

1. Social Rehabilitation

The more traditional description of rehabilitation in terms of disease, impairment, disability, and handicap has been changed due to the recent The World Health Organization's (WHO) International Classification of Functioning, Disability and Health (ICF) model providing a helpful framework for understanding of health, disease or trauma, impairment, restriction in activity and limitation in participation to activity [5]. Figure 1 illustrates a model of the sorts of interventions that could be employed to improve health, either by reducing impairments, helping performance of activities even in the face of limited capacity, or promoting participation even in the face of activity limitation. Disease is an intrinsic pathology or disorder that may affect the systems for instance, osteoarthritis, stroke, and amyotrophic lateral sclerosis. Impairment is a decline or loss in anatomic structure or function at the organ level usually as a result of the disease process. For example, decreased mobility due to osteoarthritis of the back, hips, or knees or decreased hand function due to stroke impairment predispose to disability that refer to functional limitations, that is, restrictions in the ability to carry out basic physical and mental activities for example limitations in the ability to carry out activities such as dressing or bathing. However, disability occurs due to the cumulative effects of internal factors (comorbidity, disease severity, cognition, education,

culture) and external factors (treatment facilities, environment, support services, finance). The limitation in participation, which is preferred by ICF instead of the term handicap, refers to the interaction of an impairment or disability with external factors (e.g., architectural barriers) that limit or prevent the fulfillment of a normal social role of the elderly in the society, for example having a walk in the street for shopping [3]. From this point of view, we may consider geriatric rehabilitation involving with therapeutic interventions but not only treating the elderly person or elderly patients, but also improving their overall functional capacity, preparing them to return to their active life and achieve the sustainability of the gained activities as much as possible.

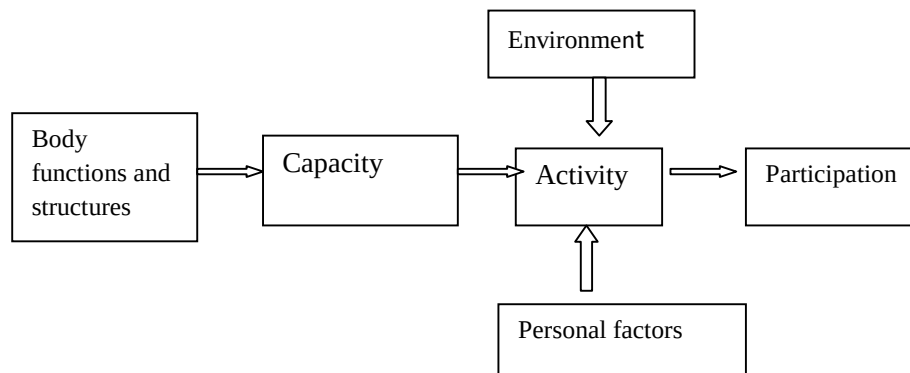


Figure 1. ICF Model of Health (Adapted from Gladman, 2008).

2. Success in Rehabilitation

Elderly who need rehabilitation interventions may have a disease or a disability from their childhood or adult life that are usually in tendency of deterioration due to the consequences of the eventual effects of aging of the systems. Thus, by years their dependency will be changed according to their needs and mostly will be increased. For instance, as a post-polio patient while walking with a long leg brace and a cane, working professionally and involved in social activities with her family and friends; towards the retirement age may have sudden symptoms of post-polio syndrome that may refrain her from having this active life and she may become more depended to home due to weakness, tiredness and pain. This situation may eventually decrease her quality of life and of course, her life satisfaction.

On the other hand, the elderly may have a healthy life without serious health problems, however, during the later life they may have a disease or a trauma that refrain them from having a healthy, active aging. Although, the degree and the duration of dependency are related with the pathology, the previous functional capacity is also an important criterion for the health promotion. Hence, the achieved personal features and the environmental facilities provided for the elderly will have a governing effect on the success of the rehabilitation. In this context, rehabilitation may be viewed as the combination of two models of health and illness—the biomedical and the biopsychosocial model. The biomedical model refers to the effects of biology, physiology, anatomy, and their clinical outcomes and focuses on function and well-being of the elderly. This model is most useful for the cause-and-effect analysis; for

example, a left middle cerebral artery stroke may cause right-sided weakness. However, the biopsychosocial model emphasizes the importance of understanding the interaction between health, illness, psychology and social factors of the individual, but not only the ongoing physical loss. Thus, the health care delivery and understanding human health and illness in their fullest contexts are essential for the quality life of the elderly. The chronic conditions that can occur with aging may force to reduce the homeostatic reserve, that is, the ability to recover from a disease or impairment. Of course, the diminished hemostatic reserve of the elderly may limit their rehabilitation potential. For example, an elderly patient with a right-sided hip and knee arthritis who requires a cane in the left hand, if he has an attack of stroke resulting with a left-sided hemiplegia may have weak rehabilitation potential [3, 7]. In this condition, the independence level and the participation of the elderly will be limited with the impairments on both sides that will both affect negatively their quality of life from the biomedical as well as biopsychosocial points of view.

Although the success of the rehabilitation program is depending on the skills and the knowledge of the team and the cooperation between the team members, it is also depending on the personal status of the elderly. These features that must be taken in consideration are classified as patient's skills gained previously (disability/activity limitations, education, socio-cultural features etc.), from the past acquisitions (active life style, sports, participation to recreative activities etc.) and from the ongoing present conditions (pathology, acute or chronic diseases, disability etc.). Thus, biopsychosocial status of the elderly individual is effective through the process of decision making on the planning geriatric rehabilitation program.

3. Clinical Assessment for Geriatric Rehabilitation

Comprehensive geriatric assessment (CGA) is a multidimensional interdisciplinary diagnostic process, focused on determining a frail elderly person's medical, psychological, and functional capabilities in order to develop a coordinated and integrated plan for treatment and long-term follow-up [8]. Geriatric rehabilitation efficiency may be improved with shorter length of stay (LOS) in the hospital, accompanied by higher efficacy scores. Elderly with shorter LOS had a relatively better rehabilitative response (better gain and efficacy scores) than those patients who experienced a longer LOS. [9]. Clinical geriatric assessment in rehabilitation is structured for the determination of the possible causes of the impairment or the disability. In this context, the assessment of a disability focus onto understanding the overall features of the patient as (1) the determination of the severity of the disability (e.g., the frequency of falls); (2) the compensatory techniques or adaptations the patient has made (e.g., use of a cane); and (3) the symptoms associated with the complaints (e.g., pain, shortness of breath, weakness, etc.)

Pain as a psychological factor that is a potentially important risk factor of disability for elderly [9]. It can be considered as a modifiable factor that negatively influences the functional recovery during the rehabilitation process [9, 10]. Musculoskeletal pain is more common and persistent among older women with disability than among men with comparable conditions and it also may increase the risk of falls [1]. Thus, the complaints about pain should be evaluated in detail in relation to its severity, location, quality, radiation, and the

timing. It is also important to question the patient about the disabilities s/he may not have mentioned. For instance the elderly who has a chief complaint of frequent falls may also likely to have difficulty such as in rising from the toilet and getting into and out of the bathtub, and so on [11]. Additionally, physical functioning issues related to walking, pain, and low endurance may be among the hidden or apparent complaints of the elderly with orthopedic conditions. Since the minimization of the loss of muscle strength and flexibility is the best way to prevent functional decline [1] the musculoskeletal assessment are in utmost importance. Apart from these concerns however, the goals, expectations, and perceived rehabilitation impact of the elderly are the other features emphasizing their performance in activities and the fulfillment of their roles in the community or in their living environment. Thus, the physical, cognitive, perceptual, and sociological functioning components that are all relevant with each other in elderly must be assessed thoroughly [7]. Table 1 shows the important rehabilitation outcomes and the examples of the relevant activities among elderly.

Table 1. Important rehabilitation outcomes and examples drawn from the older adult participants (most to least) (adapted from Demers, Ska et al., 2004)

Rehabilitation Outcomes	Examples
Recover and improve walking abilities	Walks straight, without cane or crutches; manage stairs
Maintain relationships	Engage social contacts
Reduce pain	Walk without pain; pain relief
Resume previous level of functioning	Do things as before; walk like previously
Drive and use means of transportation	Take the bus by oneself
Perform basic activities of daily living	Take care of self; eat by oneself, use the washroom; hold a cup of tea
Improve endurance	Walk for longer periods of time; not feel so tired
Perform instrumental activities of daily living	Clean the house; cook meals
Feel self-sufficient	Feel capable and independent; feel adjusted to the new life situation.
Regain confidence	Not feel so anxious; not be afraid to go outside, to use stairs; not be afraid to slip and fall
Resume social activities outside the home	Attending social events
Go out and about	Go shopping
Not to feel like a burden	Avoid having someone present at all times
Resume leisure activities	Read books; travel; practice sports)
Be independent in transfers	Transfer to the toilet; transfer into car; raise from a chair
Resume work activities and voluntary help	Help others; assist sick people
Get back home	Keep living in the family house)
Recover physical capacities	Move about; move and use arms and legs; strengthen the legs.
Recover quality of life	Maintain significant activities, not wait for time to pass.

4. Geriatric Rehabilitation for Specific Conditions

Geriatric rehabilitation covers the complex approaches of the congenital or acquired diseases or impairments of the elderly. These may be a post-operative physical therapy and rehabilitation intervention after a femoral neck fracture, cerebrovascular attack or a vascular problem; or may be an approach such as to relieve the pain, to improve the strength of the elderly who have chronic musculoskeletal or neurologic problems. However, the elderly may

have combination of the different impairments and sicknesses such as diabetes, hypertension, incontinence that may deteriorate their quality of life drastically so that the rehabilitation program should be planned accordingly. However, besides the variety of different medical conditions, we intended to discuss the fundamental approaches as orthopedic and neurologic geriatric rehabilitation that may be common among the elderly living in the society or the institutions in this chapter.

4.1. Geriatric Orthopedic Rehabilitation

Orthopedic rehabilitation is based on a sequence of events proceeding from the acquisition of pathology (acute or chronic diseases, or injury) to the generation of impairments manifested at the level of organs and body systems (e.g., a decrease in muscle strength, flexibility, aerobic capacity and cognitive capacity), which leads to functional limitations at the level of the organism as a whole (e.g., restrictions in walking, stair climbing, general mobility skills and balance), and finally to disability (e.g., difficulty or inability to manage daily activities of personal care, household chores, and employment and leisure activities within the context of the environment and its challenges) [12, 13].

Osteoarthritis

Osteoarthritis (OA) as a highly prevalent, progressive, degenerative disease is causing functional impairment and/or disability [14-17].

It is associated by pain, impaired muscular stabilization and reduced range of motion, and the progressive loss of functions [18, 19].

However, flexibility exercises are important to decrease stiffness, increase joint mobility, and prevent soft tissue contracture. Strength training is an essential part of rehabilitation for people with osteoarthritis. Resistance training may reverse many age related physiological changes and can enhance the function by improving the strength of muscles that supports affected joint. Isometric exercises are preferred improve muscle strength and static endurance and are better options if the joint is inflamed or unstable [20, 21].

Osteoporosis

The physical inactivity/sedentary life style and impaired neuromuscular function are among the risk factors of osteoporosis besides the genetic features and smoking [22]. A negative correlation has been reported between the strength of back extensor muscle and thoracic kyphosis in the women with osteoporosis. It was showed that the relative risk for compression fractures was 2.7 times lower in women (56 years and above) trained with the extensor exercises compared to control group. Specific balance training can also be helpful to manage the osteoporosis among elderly, because patients with osteoporosis and kyphosis maintain their balance with an increased use of hip joints movements [21].

Hip Fracture

Hip fractures are usually associated with osteoporosis and a fall [22, 23]. Most hip fracture patients have difficulty in standing, and the involved leg appears to be shorter and externally rotated. Residents of nursing homes and long-term care facilities are at high risk for fractures. Most have low Body Mass Density (BMD) and a high prevalence of other risks

for fracture, including advanced age, poor physical function, low muscle strength, decreased cognition, and high rates of dementia, poor nutrition, and multiple-drug regimen [23].

Table 2 shows the negative predictors affecting the post-operative *recovery* process of the hip fracture. Due to the prolonged limited mobility during the postoperative course of the hip fracture deconditioning is common among elderly. The deconditioning treatment includes regaining the upright posture, improving joint ROM (range of motion) with active or passive exercises, increasing strength and coordination with the combination of isometric, isotonic, and functional activities. These interventions should be combined with sensory and intellectual stimulations to provide the cognitive improvement as well [24]. Reduction of modifiable risk factors is important to prevent and decrease fracture risks [22]. Physical activity such as resistance exercise training and weight-bearing exercises may have a two-fold contribution to reduce fracture risk. Low-impact exercises, such as walking, have beneficial effects on other aspects of health and function. Hip protectors as hard or soft shells with soft padding cover the area over the greater trochanter of the hip joint and can reduce the risk of hip fractures in nursing home residents [23].

Table 2. Post-Hip Fracture Surgery: Negative Outcome Predictors
(adapted from Masqueda, Chinh 2004)

Poor mobility on the second day, post-operatively
Old age (>75)
Use of a gait aid before fracture
Poor mental status
Nursing home resident
Poor grip and ankle dorsiflexion strength
Significant number of comorbidities
Limited number of leisure activities, poor coping skills
Hip pain, poor self-rated health

Table 3. Effects of the Aging Process on Selected Physiologic and Health- Related Variables (adapted from American College of Sports Medicine. ACSM's Guidelines for Exercise Testing and Prescription, 2006)

Variables	Change
Resting hearth rate	Unchanged
Maximal hearth rate	Lower
Maximal cardiac output	Lower
Resting and exercise blood pressure	Higher
Maximal VO_2 ($\text{L} \cdot \text{min}^{-1} \cdot \text{mL} \cdot \text{kg} \cdot \text{min}^{-1}$)	Lower
Residual volume	Higher
Vital capacity	Lower
Reaction time	Longer
Muscular strength	Lower
Flexibility	Lower
Bone mass	Lower
Fat-free body mass	Lower
Percent body fat	Higher
Glucose tolerance	Lower
Recovery time	Longer

Table 4. Geriatric Rehabilitation Program of Elderly with Neurologic Problems

Aim	Intervention
Improving motor control	Neurofacilitation Technique: Bobath, Brunnstrom, Rood, PNF ¹ techniques, FES ² , biofeedback
Chest physiotherapy	Respiratory control, respiratory exercises Coughing and force expiratory exercises Bronchial hygiene and its training Mobility exercises
Balance and weight-bearing exercises	Neurofacilitation Techniques Mat exercises, exercises in-and-out of the parallel bar Proprioceptive exercises Balance and weight-bearing exercises in ADL ³
Prevention of trunk control	Shoulder girdle and pelvic control Trunk and abdominal strengthening and proprioceptive exercises Improving the body awareness
Gait training	Characteristics of gait (style, rhythm, speed, step number, step length etc.) Exercises for the asymmetry of trunk Kinesthetic exercises for step and breath control Functional mobility exercises
Fall prevention	Strengthening of pelvic muscles (hip flexors, M. Gluteus maximus and medius), ankle joint muscles (M. Tibialis anterior and peroneals), M. Tensor fascia lata) Appropriate shoe for in-and-outdoors Home modifications Shock absorbent pillows, carpet softening layers Gait training
Orthotic approaches	Positioning splints (antispasmodic splints for hand, foot) AFO*(night/day), UCBL** foot orthosis, Shoe modifications
Edema control	Elevation, lymphatic drainage

¹ Proprioceptive neurofacilitation, ² Functional electric stimulation, ³ Activities of Daily Living

* Ankle Foot Orthosis, **University of California Biomechanics Laboratory

4.2. Geriatric Neurologic Rehabilitation

Congenital or acquired childhood diseases as cerebral palsy (CP), spina bifida, poliomyelitis; adult life diseases as multiple sclerosis, Parkinson's disease, amniotrophic lateral sclerosis or diseases that can be seen in any age as stroke or spinal cord injuries result with increased physical, mental, psychosocial consequences of aging. Dementia, especially Alzheimer disease conduce physical disabilities that result with dependency to bed. Thus, the geriatric neurologic physiotherapy and rehabilitation approaches are important to provide an improvement in function or to delay the eventual deterioration of elderly due to the ongoing pathology (Table 4). However, the interest of patients having chronic neurological diseases since their childhood to regular physiotherapy and rehabilitation interventions and follow-ups are gradually decreasing as they step into adult life and as they become elderly. This seems not quite different among their families as well. In this context, we may conclude that the secondary problems of aging have tendency to be enhanced and the overall quality of life of the elderly having neurological problems since their childhood may gradually be diminished [25].

Geriatric Rehabilitation in Congenital or Acquired Childhood Diseases

An elderly person with *cerebral palsy (CP)* faces with physical changes in their functionality, especially regarding to their mobility besides the physiological changes occurring due to aging. The contractures and deformities may tend to increase the severity of their functional limitations as they become elderly. The knee and foot deformities as well as the hip dislocation, scoliosis and the pelvic obliquity are the most common conditions tend to increase, that all require special care of physiotherapy and rehabilitation, mainly for hygiene, positioning and transfers [21, 25]. The pain (mainly on hips, knees, feet, lumbar and cervical spine) is also another feature that should be considered in geriatric neurorehabilitation of persons with CP. Thus the family and the care givers should be observant about the possibility of pain, even if, they do not complain, especially for the ones with communication and cognitive problems [26]. Increased weakness of the muscles due to aging may cause bowel and bladder incontinence that limits their activity of daily livings and of course, their quality of life [27, 28]. Additionally, elderly with spastic or dyskinetic CP may have oromotor problems, chewing, swallowing and eating disorders. This may cause inefficient nutrition and malnutrition, as well as increased risk of aspiration [29]. Hurvitz et al. (2008) have reported high incidence (40%) of underweight due to inefficient nutrition among non-ambulatory children ($n=137$; 9.8 ± 4.7) with cerebral palsy. However, in this group of children, they have also found that overweight and risk of overweight were 33% and 21% among the ambulatory and non-ambulatory ones, respectively [30]. Thus, obesity is another problem due to the limited physical activity level of children with CP that may have tendency to increase by aging, especially if they have cognitive disabilities, as well [31]. The physical therapy and rehabilitation program for the elderly with cerebral palsy need to be designed after a thorough assessment (Table 5). The rehabilitative program should emphasize on preventive approach and consider mainly the proper positioning and the prevention of mobility. If the elderly have cognitive disabilities the caring team should be careful on the positions or activities that may cause or alleviate pain. The daily living activities during a home or an institutional care are also covering the important portion of the rehabilitation program. The involvement to the recreative and leisure events could also be encouraged even if the elderly may not have any previous habits.

Table 5. Assessment in Neurologic Rehabilitation

1- History and sociodemographic features and environmental factors (age, marital status, life style before the disease, living environment, the aspects of the patient and the family to stroke and rehabilitation, their expectations, support of family and relatives, where to live after discharge)
2- Motor function assessment (Motor Assessment Scale, Motor Function Measure),
3- Balance (Berg Balance Assessment)
4- Mobility (Rivermead Mobility Index)
5- Activities of Daily Living Assessment (Barthel Index, Functional Independence Measure - FIM)
6- Neurophysiologic approaches (Rood, proprioceptive neuromuscular facilitation-PNF, Brunnstrom, Bobath, Margaret-Johnstone Techniques)
7- Muscle strength and impairments (gross motor test, manual muscle test- MMT, myometric tests, grip power, pinch power test)
8- Range of motion (goniometric tests)
9- Muscle shortenings and deformities (hand deformities, knee flexors contracture, Equino varuos)
10- Respiration (frequency, depth, type-diaphragmatic or chest respiration)
11- Pain (McGhill pain test, visual perception scale)
12- Skin (pressure sores) and sensation (touch, hot-cold, point-dull)

If elderly with poliomyelitis have a *post-polio syndrome (PPS)* should be suggested to change their life style not to alleviate the symptoms as weakness, fatigue, and pain that may arise with physical activity, especially as the day progresses. Thus, the physical therapy and rehabilitation program should be planned according to the physical condition of the patient. However, the manifestations of general fatigue may result with increased sleep requirements and decreased concentration that the physiotherapist or the occupational therapist should take these in consideration while planning the treatment, as well as the activity program [32].

Thus, while planning the rehabilitation interventions and the exercise treatment programs, exercise items should be;

- Given in low intensity and low repetitions,
- Distributed in the day time when the elderly not feeling tired, mainly in the morning hours.

They are also suggested to adjust themselves to a more economic functional level for not to alleviate the general fatigue and weakness with long and often rest intervals between the activities as well as the exercises. The utility of the wheelchair during the day for a few hours when the elderly feel tired can be considered, even if they do not need according to their functional level of independence.

Geriatric Rehabilitation in Stroke

Stroke as a disease causes long-term disability in any age of life may result with more serious disabilities in elderly age. However, elderly having stroke weather in young or adult ages or in later ages, unfortunately, usually have less attention by the medical team regarding to their rehabilitation process. They are either discharged home after a short period of rehabilitation program or if they cannot live at home, are discharged for a nursing home. Although elderly care in nursing homes are very well organized, the inefficiency in rehabilitation programs are often common in institutional settings due to the administrative structures (such as, lack of time or staff, changes in treatment time table) or due to the features of the elderly person (such as decreased motivation, changes in physical features or environment). In this context, it is generally a common thought that the institutionalized elderly with stroke may have tendency to lose their gained activity level in the acute and subacute rehabilitation programs administered in the hospital [33-35].

The main goal of geriatric rehabilitation of the patients with stroke is to achieve the most possible independent condition, physically, mentally and sensorial. Thus, the necessary compensation mechanisms are to be furnished in the area of supportive devices such as, orthoses, canes, walkers, wheelchairs etc. (Figure 2) and assistive devices such as touch screens, ergo keyboards or pointing devices. However, aging negatively affects the outcome measures of the rehabilitation program, especially for the ones over 85 and more [33].

Geriatric Rehabilitation in Spinal Cord Injuries

Spinal cord injuries either occurred in early or in advanced ages result with decreased quality of life and limited social inclusion of the elderly. This may be due to the decline in



a



b



c

Figure 2. A cane (a), Canadian crutches (b) and a wheelchair (c) for balance and mobility to provide ambulation of the elderly (b).

physical, cognitive and sensorial conditions and also the consequences of the cardiovascular or respiratory diseases that are commonly seen in advanced ages. [36] reported that the incidence of wheelchair usage and dependency in activities of daily living as well as the bowel and bladder incontinence are higher among the elderly than the younger ones having

more or less the same level of spinal cord injuries. They may also be affected much more seriously from the bed sores and the disturbances of the venues return that these features should be taken in consideration as a part of the rehabilitation program. Accordingly, besides the individualized rehabilitation programs aiming to maintain the physical condition of the elderly, group therapy programs emphasizing on the activities of daily living, recreation and leisure improve their independence level and social interactions.

Geriatric Rehabilitation in Diseases of Older Ages

The aims of the rehabilitation program of the person with *Parkinson's disease*, *Multiple Sclerosis* or *Amyotrophic Lateral Sclerosis (ALS)* that are mostly seen in older ages, include generally to prevent and upgrade their functional level and the quality of life as much as possible. However, according to the special characteristics of the diseases, there are certain treatment approaches that need to be emphasized on during the regular physiotherapy and rehabilitation program. For instance, the balance and co-ordination exercises are important in the rehabilitation of the elderly with Parkinson's disease to achieve better equilibrium in the movements of the body, not only for the lower extremities for better gait cycle and for prevention of falls, but also for the upper extremities for daily living (as basic, instrumental and advanced activities of daily living-ADL), recreative and leisure activities. Resting periods between the exercises are important during the rehabilitation program of the patients with multiple sclerosis; especially the afternoon fatigue should be taken in consideration by the physiotherapist. Although the respiratory physiotherapy is fundamental part of the geriatric rehabilitation, it is vital for the ALS patients as well as swallowing treatment and training of morsel control (Table 6) [21, 25].

Table 6. Swallowing treatment and morsel control

Aim	Intervention
Provide head and neck muscle control	Strengthening and flexibility exercises for upper trunk, head and neck region
Improve control of face and mimic muscles	Strengthening exercises for mimic muscles (isotonic, isometric, PNF ¹), blowing (feather, paper, pipette, whistle etc. vb.), draw in-out (having fluid with pipettes in different diameters) Combined exercises of head and neck with mimic muscles
Posture exercises	Postural exercises in different positions
Provide respiratory control	Respiratory exercises (diaphragmatic respiration, thoracic mobility and expansion exercises)
Provide morsel and swallowing control	Swallowing training of the patient and care givers (small morsels wetted with liquid, chewing well, drinking thickened liquid, water thickened with special gel, not to have water before swallowing the morsel, drinking liquid with pipette or baby glass with little holes, swallowing while flexing the head)

¹ Proprioceptive neurofacilitation

4.3. Geriatric Rehabilitation in Dementia

If elderly is diagnosed with dementia, special care must be given during the rehabilitation program regardless of the degree of dementia. The physiotherapy and occupational therapy

are among the effective approaches to decrease the severity or to delay the physical and social consequences of dementia [37-39]. Thus, while planning the exercise programs to improve the functional levels of elderly with dementia, the specific approaches should be focused on to support them cognitively and to foster their memory on the given tasks and/or on the previously learnt activities. In this context, the motor learning procedures can be considered as effective tools of the cognitive approaches in geriatric rehabilitation [40].

Motor Learning Approaches in Dementia

Although different variety of techniques in teaching a task is available in motor learning [41], the most suitable one for geriatric rehabilitation is teaching the task from part to whole by dividing the task into steps and making it as simple as possible. Then, the main goal is to reach the whole of the task through combining the steps, until the integrity of the task is achieved in elderly as a learnt activity. For this purpose, variant reinforcement means as clueing and prompting (that both can be both verbal, visual, physical or the mixture of all), imagining the task, and cueing the parts of the task in different aspects could be utilized by the physiotherapist. Several repetitions of each step is necessary for adapting the activity. Then via combining the steps with feedbacking the previously learnt ones until the whole of the task is achieved skillfully so the task can be executed whenever it is needed such as during an exercise session or during an activity of a daily life or recreation. Another words, the task is to be practiced efficiently to achieve a sustainable learnt activity for the elderly [40, 41].

However, depending on the cognitive level of the person the planned goals in realizing the task may have different diversities. It may not be possible to reach to the final task due to the cognitive capacity of the person [33, 38]. Then the program should be modified according to the needs of the person. For instance, if the planned task is washing hands or brushing the teeth in the morning, picture clues can be stricken by the lavatory on the wall or mirror, and if needed by the soap for a patient with dementia to remind this. Of course, this may not be efficient and the verbal or physical prompts may be needed to support the remembrance of the elderly. On the other hand, the safety precautions are also important, such as, against the tread of using hot water in the tab; the water temperature can be fixed in a warm level in advance to prevent burns.

Activities of Daily Living Training for Elderly with Dementia

During the exercise program, the skills of the activity of daily living, especially the instrumental type such as the ability to use telephone, shopping, housekeeping, laundry, food preparation etc., and the mobility at indoors and outdoors are aimed to be improved. [28, 29, 33]. However, these are embedded with exercises to support or stimulate the cognitive functions and abilities of the elderly for example on their memory, attention, the visiospatial functioning, reasoning and learning-recalling the new information skills [25, 40]. During these exercises their cognition can be stimulated through their possibly diminished executive functioning skills as planning, judgment and insight. For this purpose, the exercises are assessed according to their aims and consequences, by means of providing the media for elderly to find a relation between the reason and the result of the tasks. For example, during the dressing exercises, the elderly can be encouraged to find out the reason of the task that is to *put on a sweater*, and then, the result of it that is *for not to have a cold*. On the other hand, more sophisticated exercise patterns can be preferred by the therapist according to the need of

the elderly, such as they may be guided in selecting one of the tasks; deciding on the next task can be done; deciding on a task for her/his partner during the activity; selecting an exercise for her/his partner or for the group in a group treatment; selecting an exercise according to the day or the time, such as deciding to have a walk by the sea in a nice weather or to have an indoor exercise program in a cold weather. The elderly may be involved much more actively into the exercise programs by having the responsibility of counting the repetition by grouping the tasks according to the number of repetitions, the affected area on body or the features of the exercises as strengthening, stretching, and coordination and so on. Then, one more step further in exercise regime is to demand from the person to remember, define and execute the task for example according to the planned schedule written on the board or on the refrigerator.

Overall, the variety of different mental activities can be selected by the therapist to guide the person to fulfill the planned functions. However, the family and the caregivers should be included into the practices and they must be aware of that many repetitions and trials are needed for the success. Thus, the patience and motivation of the medical team, the elderly person, her/his family and the caregivers are noteworthy.

4.4. Balance Problem and Fall Prevention in the Elderly

The risk of falling and suffering injury increases dramatically especially in elderly. In frail institutionalized populations, almost threefold higher incidence of falls has been noted. Even falls which result in no physical injury often have serious social and psychological consequences [42].

It was implied that 40% of 90% ambulatory nursing home patients fell two or more times in the previous 6 months. One in 10 falls results in serious injury and is the cause of mortality in 40% of injury- related deaths in adults over age 65. Serious injuries commonly resulting from falls include fractures of the femoral neck, limbs, and vertebrae, and subdural hematoma [43]. With advancing age, changes related to normal aging, and those associated with diseases and their treatments, can affect the integrity and function of the musculoskeletal, vestibular, central, and peripheral nervous systems. These effects are further compounded by a reduced capacity for plasticity and repair in the elderly [44]. Postural insufficiency, gait disorders and fear of falling have been observed after falling in frail older patients [42]. Remaining on the ground for more than one hour after a fall is a marker of muscle weakness, illness, and social isolation in the elderly that increases the outcome effects of the fall. They may experience decreased confidence in their physical capabilities and reduced physical activities. Thus, in geriatric rehabilitation the main goal is actually the prevention of fall that is depending on many factors, including both extrinsic and environmental factors and intrinsic factors or personal factors. Although some falls may have a single cause, most falls are believed to result from a combination of factors. Not all risks can be eliminated, but the modification of even one risk factor can be a worthwhile therapeutic goal in geriatric rehabilitation [43, 45, 46]. For example, strengthening exercises for the antigravity muscles, stretching exercises for the shortened tight hamstring muscles may increase the confidence of elderly while walking or stair climbing.

For a comprehensive approach to elderly in rehabilitation a thorough physical assessment including examination of strength, power, balance, posture, and gait is essential. These assessments also include evaluation of the coordination of general body, as well as the vision,

hand and leg movements, sensation and proprioception. The gait assessment tests evaluating the dynamic balance and the quality of walking, including normal gait, tandem walking as well as dual task activities, such as walking while carrying an object or head movements during walking are to have a detailed information about the elderly. Several performance-based assessments of daily living activities and computerized dynamic posturography tests are also to understand their physical capacity and balance quantitatively [44].

4.5. General Exercise Prescription for the Elderly

The skeletal muscle strength begins to decline at approximately age 45 and is associated with a 30% to 40% decrease in strength by the age 80 years. This loss of muscle mass not only causes a reduction in strength but may be responsible for as much as 30% of the decline in maximal rate of O₂ consumption (VO₂max) [39, 47]. The loss of skeletal muscle mass below a critical threshold (sarcopenia) leads to a functional impairment and frailty. As the population surveys suggest muscle loss is common and highly correlated with the decreased function in older adults. Approximately 8% of men and 10% of women had values of muscle mass below two standard deviations of young adult values (class II sarcopenia). It has been established that sarcopenia, characterized by diminished muscle mass, strength, and power, is a key common denominator in the development of frailty [48, 49]

Regular participation in physical activity and/ or exercise is not only integral to the maintenance of good health and functional independence in older adulthood, but also serves as a primary role in the prevention of numerous chronic diseases (e.g., type 2 diabetes, cardiovascular disease, osteoporosis, certain types of cancer) and cognitive decline [48, 50]. Conversely, inactivity doubles the risks of developing a disability that will adversely affect mobility as well as the ability to perform even the most basic activities of daily life. This downward spiral in physical function ultimately heightens the risk of both the loss of functional independence and falls [46, 51].

It has been identified that the general body exercises have a potential to influence on functional balance and mobility in geriatric inpatients. It is possible that these frail elderly would have benefit from the resistance exercises for the leg muscles [52]. However, ankle exercises directed at increasing ankle ROM may have an influence on reducing falls in this population [53]. Additionally, it has been concluded that the ankle ROM exercises have potential increase on the effectiveness of clinical and community interventions designed for improving balance and reducing falls in elderly women.

Although the general principles of exercises and their relative adaptations are similar in different age groups, there are particular differences especially for the exercise programs aiming to improve cardiorespiratory fitness, flexibility as well as resistance training [54].

In this context, it is important to know whether the elderly is having a systemic disease or not for the physiotherapist during planning the physiotherapy program. Thus, to design a safe and an effective exercise prescription for elderly, the physiological changes in aging as well as the possibility of the active or latent processes of the systemic diseases should be considered (Tab. 3). On the other hand, the necessary consultations should be realized before the initiation of the treatment program [54].

Resistance Training for Elderly

Resistance training has potential to increase muscular strength, power, and endurance of elderly, improve mobility and prevent falls and fractures [54, 55]. Thus, resistance training is effective, especially, on frail elderly to improve their physical fitness that the followings are the guidelines of resistance training recommended for elderly [54, 56]. The resistance training sessions should be closely supervised and monitored, especially the first a few sessions until the elderly is acquainted. It should begin with minimal resistance to allow for the adaptation of the connective tissue elements. The followings are the important features of a resistance-training program planned by the physiotherapists for elderly.

- The exercises should be performed as a set of 8 to 10 exercises mainly for all the major muscle groups.
- A set of exercise should involve 10-15 repetitions that elicit a perceived exertion rating at the level of 12 to 13, which corresponds to “somewhat hard”.
- The multi-joint (as opposed to single-joint) exercises should be performed.
- The exercises should be performed in range of motion that is within a “pain-free arc”, which means the maximum range of motion that does not elicit pain or discomfort.
- First the number of repetitions should be increased then the amount of resistance for achieving a safe overload.
- When returning from a layoff of more than 3 weeks, it should be started with the resistance of 50% or less than the previous training intensity, and then the resistance should be increased gradually.
- The participants should be instructed to maintain their normal breathing pattern while exercising.
- Routine activities (e.g., domestic work, gardening, walking) may help to maintain muscular strength.

Flexibility Exercises for Elderly

A well-planned program of stretching can counteract the usual decline in the flexibility of elderly people, may improve balance and agility. Thus, the flexibility exercises have potential to enhance their functional capacity and reduce the injury potential (e.g., risk of muscle strain, falls). Exercises should be planned for the major joints in the body and for the back, upper trunk and neck region with mild stretching [54, 55]. Yoga and tai-chi may be helpful in improving or maintaining the flexibility of the body [57] that could be included into the rehabilitation program. In this regard, a general exercise prescription for achieving and maintaining flexibility of elderly may include the following approaches [54, 58].

- The flexibility exercises should be performed a minimum of 2 to 3 days/week,
- Stretching should be performed till the end of the range of motion at a point of
- Tightness, however, without inducing any discomfort,
- Performing static stretching exercises routinely for the major joints and the muscle
- Tendon units to increase the range of motion.
- Holding the each stretch for 15-30 seconds to achieve the muscular adaptation [58].

Mode of Exercises

Although there are variety of exercise models as isometric, isotonic (eccentric, concentric) exercises for elderly, they should be selected specifically for not impose excessive orthopedic stress to the body. Isotonic exercises are preferred according to the needs of the elderly as passive, active, active assistive or resistive modes. However, the outdoor walking is an excellent mode of exercise for many elderly people that could be suggested safely. As an effective weight-bearing exercise it is also important for the psychosocial status of the elderly since they are naturally encouraged for the social interactions while walking outdoors with their peers or friends. Swimming, aquatic exercise and stationary cycle exercises may be especially advantageous for those with reduced ability to tolerate weight-bearing exercises [48, 54].

The exercises should be accessible according to the physical and mental conditions of the elderly; however, they should be enjoyable as well, to improve their exercise habits. However, a group setting such as callisthenic exercises may provide important social reinforcement to improve their exercise adherence. Hence, if possible, the group exercises could be embedded into the exercise rehabilitation programs to challenge their participation [54, 55].

Intensity of Exercises

It is important to remember that, the exercise intensity for the active elderly people should start at low level and gradually may progress according to their tolerance and preference individually for not to alleviate any medical problem and also for the promotion of long term fitness. For instance, the measured peak hearth rate is preferable to an age predicted peak hearth rate when prescribing aerobic exercises because of the variability in peak hearth rate for the individuals above 65 years of age and also higher risk of underlying coronary disease [54, 55].

Thus, initiating a program at less than 40% VO_2max is not unusual. The exercises need not to be vigorous and continuous to be beneficial; a daily accumulation 30 minutes of moderate intensity of physical activity may efficiently provide health benefits [39, 47, 54].

Duration of Exercises

Exercise duration need not to be continuous to produce benefits, thus, those who have difficulty sustaining exercise for 30 minutes or who prefer shorter bouts of exercise can be advised to exercise for 10 minute periods at different times throughout the day. It is preferable for the elderly to increase exercise duration rather than intensity to avoid injury and ensure safety [47, 54]

Frequency of Exercises

Physical activity performed at moderate intensity should be performed most days of the week. If exercise is undertaken at a vigorous level, it should be performed at least 4- 2 days/ week or at low-to moderate intensity on alternating days [54, 55]

4.6. Exercises in Special Conditions in Geriatric Rehabilitation

The benefits of exercise rehabilitation as improving joint mobility and stability, musculoskeletal strength and endurance; enhancing neuromuscular coordination and agility as

well as functional capacity and cardiovascular fitness also include rising the psychosocial status of the elderly [51, 54]. As exercises play a fundamental and invaluable role in rehabilitation programs, they must be designed and implemented to meet the specific needs of the elderly patients.

Exercises for Fall and Balance

It was documented that frail elderly people with a history of falls must benefit from vestibular rehabilitation programs in order to avoid the loss of functional capacities and disability. These specific rehabilitation programs are based on the basic motor patterns of training, including the practice of vestibular and balance exercises as well as gait training. This program can be intensified through a multidisciplinary approach, which requires the education of all the members of the team taking care of the elderly [42].

Vestibular rehabilitation is designed to restore homeostasis of the balance system. The goals of vestibular rehabilitation include decreasing dizziness, improving balance, and minimization the fall risk by improving vestibulo-ocular reflex accuracy, postural control, and oculomotor skills. Various techniques used in vestibular and balance rehabilitation therapy include adaptation exercises, habituation exercises, substitution exercises, balance exercises, gait and general conditioning exercises. The planned exercise program should be consistent with the patient's needs, so that it is specific for the each particular case [44].

It was demonstrated that the use of other forms of exercises such as Tai Chi is a good evidence as a medium for lowering fall risk among more sedentary community-residing elderly [46]. Low, Ang et al. (2009) have reported in their systematic review that Tai Chi has the potential to reduce falls or risk of falls among the elderly, if they are relatively young and non-frail [57]. The longitudinal study on the effects of Tai Chi Exercise among elderly with osteoarthritis (OA) suggested that Tai Chi should be encouraged to maintain physical function and improve quality of life elderly people with OA [59].

Cardiorespiratory Fitness Exercises for Elderly

Elderly people should be encouraged to do for at least 30 minutes of moderate- intensity physical activity on most and preferably, all days of the week. This can be accomplished with activities such as brisk walking, gardening, yard work, housework, climbing stairs, and active recreational pursuits. For those achieving this level, additional benefits may be obtained with longer – duration moderate – intensity physical activity or by substituting moderate with higher intensity physical activity. It is important to remember that the activities performed at a given Metabolic Equivalent Threshold (MET) value may represent greater relative effort in elderly than young people because of the decrease in peak MET's with age. On the other hand, the optimal mode of exercise for elderly can be influenced by the physiologic and psychosocial variables, such as work capacity, orthopedic problems, poor balance, and travel limitations [48, 54, 55]. Thus, while planning an exercise program for the elderly, the mode, intensity, duration and frequency of the exercises should be specific in relation to their physiological and psychological conditions.

Exercise Training for Elderly with Dementia

In this perspective, the exercise programs for elderly with dementia need to follow the motor learning approaches [41]. The wide joint movements of proximal joints according to

the need of the elderly can be planned as active, active assistive or passive modes. If the elderly is having difficulty to perform the planned exercise then the motor learning approaches can be introduced. The observation of the elderly is important to understand the type of reinforcement they need. For instance, if the elderly is able to follow and repeat what s/he has seen, then the physiotherapist may prefer to do the assigned exercise and expect the elderly to follow and repeat the same task. However, if s/he cannot manage the task as needed, then the physiotherapist may give verbal clues, such as “You need to bend your elbow more!” or “Just bend your elbow more! If this is not efficient then the physiotherapist can show the movement on the elderly physically through passive and/or active assistive repetitions, and expect the elderly to realize the task by her/his own. Then verbal and visual prompts can be used to inform the person that s/he is successful in performing the task for the motivation. These may be visual prompts such as just an affirmation by head or hand gesture or by an applause; verbal prompts such as “Yes, very well!”, “Good, go ahead! As the elderly is able to do the task, gradually these supports should be withdrawn and expect her/him to do the task independently. Then the next step of the exercise, for instance, repeating the exercise with resistance given by weights attached to the extremities [49, 56].

On the other hand, if possible, the distal joints as wrist and fingers can be involved first actively, and then resistively. After the elderly is able to hold the weights or the elastic bands against the gravity during the arm movements, the combined arm movements that are used in daily living activities can be practiced.

The importance of weights is of course to increase the muscle power and the bone strength, but also is to provide a good base for the functional daily living activities. Weights may enhance the strength of the muscles controlling the distal joints and provide a firm base for the skilled activities that will improve dexterity [50]. For instance, a dumbbell with half a kilogram bilaterally firmly held by the elderly with dementia during the combined shoulder and elbow flexion and extension movements, could be a good sign that s/he will be able to carry an object securely or hold a glass of water and drink it with shoulder and elbow flexion (including of course slight shoulder abduction and internal rotation with forearm pronation that need to be worked on, if needed) without forgetting to hold and dropping it down.

Thus, according to the Magill's (2004) motor learning concept, the exercises planned for the elderly with dementia is to be first simple and for large joint movements. Then, as the progresses are achieved, it can be continued towards the more complicated ones; such as exercises of the distal joints can be involved that are mostly requiring skills and dexterity. For instance basic or instrumental activities of daily living, arts and crafts, ball games walking in different speeds or on different surfaces etc. [41].

5. Assistive Technology in Geriatric Rehabilitation

Assistive technology helps elderly to live more independent lives in their communities by minimizing the need for assistance from other people as well as their disability. Assistive technology devices can indeed make the task to be accomplished easier, such as, a cane or a walker as a walking aid, a prosthetic leg for an elderly amputee or a wheelchair for an hemiplegic elderly. These examples are simple and quite common assistive devices to perform the daily living [60, 61]. However, the engineers, bioengineers or designers to

improve the advanced, instrumental and/or basic activities of daily living can realize special modifications regarding to the need of the elderly and technological improvements. For instance a cane has a lamp switched on by the elderly to lighten the gloomy corridors or the stairs can be quite handy. Additionally, the canes illuminating instantly as the environment is getting darker are useful for the elderly with cognitive problems who may not be able to remember to switch on [62]

Assistive Devices as Mobility Aids

The mobility aids as canes, Canadian crutches, walkers etc. are mainly preferred for function, balance and for the activities of daily living [63]. They should be selected specially for the elderly to achieve the properly planned benefits, however, if somehow, they are not appropriate for the needs of the elderly, then either they may not have any effective support or may not be accepted. If the motivation of the elderly towards the mobility devices is decreased, they may not be used properly or they may be abandoned on a shelf or in a balcony at home, uselessly. In the institutional settings, the elderly may be motivated to use their mobility devices much more effectively by the professionals than their family members or the caregivers. Additionally, the request of modifications of the doctors, physiotherapists or the nurses for better alignment, fitting or repairments of the aids may be fulfilled much more easily and faster in the institutions. However, the caregivers may intend to utilize these devices inappropriately for other elderly to save time or expenses that may be harmful for the elderly since they were not designed for them.

Assistive devices for physical function are classified according to their indications, materials used and the technique of the manufacturing. However, they must be lightweight and comfortable, and must not require excessive force for the elderly. They must be easily done and off, and they must not be costly.

Since their aim is to increase the functionality of the elderly, Spirduso et.al (2005) have suggested to evaluate the elderly in their daily living activities as physically elite elderly or an elderly with disability regardless if they are using an assistive device or not. These devices may be a cane, a Canadian crutch, an ankle foot orthoses (AFO) or a prosthesis according to the needs of the elderly. However, mostly they may need to use more than one device in their functional activities for instance, a cane and an AFO may be essential for the elderly with stroke for walking, however, they may need wheelchairs either for transferring in longer distances and/or for during certain times of the day when they feel tired [63]. Hoenig, (2004) searched the assistive devices used by the elderly as mobility aids in United States between the years of 1989-1994. He reported that 52.1% of the elderly were using a lower limb orthoses, 82.6% were using a wheelchair and 70.15 were having a walker for their transfers. Additionally, one over three people was using more than one orthosis together. He had reached to the conclusion that unfortunately these elderly had inadequate information regarding to the effects of their mobility aids, and their financial resources were limited to have the required aids. He had also reported that the families and the medical team complained about the costliness of the orthoses, as well as the incompetence of the manufacturers and the misuse of the sources [60]

Walking Aids

Elderly may need a *cane* for walking, standing up or standing regardless the pathology. Therefore they need strong upper extremity muscles in the opposite site of the weak lower limb. The effected leg is supported through the cane on the opposite site that can cover almost 15-20% of the weight bearing. For the effective use of a cane, the elbow and wrist extensors and shoulder muscles of the elderly should be strong enough to support the body. Hence, the followings should be considered for a proper use in a thorough assessment of the elderly and the cane;

- The length of the cane must be appropriate to the arm position and the leg length of the elderly,
- Normally, when the cane was hold, the first is to be at the level of the greater trochanter while the elbow joint is in the neutral carrying angle (20-25°),
- The tip of the cane is located lateral side of the foot, 10-15cm away from the fifth metatarsal head.
- The rubber tip of the cane can be in vacuum shaped to provide much more stable hold to the floor and avoid sliding.

Weight-bearing area is increased with the help of a cane to support the body to increase the balance. However, if a wider support is needed, then a tripod or a quad cane may be preferred (Fig 2). Elderly with visual problems may also need to use cane to have information about ground through the sensorial information received by arms.

If both of the legs need to be supported, bilateral *Canadian crutches* or *walkers* can be used as walking aids. The forearm supports of the Canadians should end almost 7-8cm below the both olecranons for a comfortable weight bearing. If the elderly is using a walker, the height is at the level of trochanter majors on both side and it should have a well-balanced legs. They can be preferred for the elderly with weak upper extremities and inefficient co-ordination. Obesity can be another indication for a walker instead of Canadians. There are different types of walkers with wheels, seats, baskets etc. to easy the usage for elderly in outdoors. The elderly with Parkinson's disease may need walkers with wheels and brakes to guide patient and help to initiate or stop the movement [60].

Orthoses

Orthoses, such as Ankle Foot Orthosis (AFO), Knee Ankle Foot Orthosis (KAFO), Hip Knee Ankle Foot Orthosis (HKAFO), and University of California Biomechanics Laboratory (UCBL) are used daytime for standing and walking and night time for positioning (AFO) the extremities. However, in some conditions the modifications of the orthoses may be necessary according to the needs of the patients these may be a heel rise or a T band or both as shown in figure 3 that is used to be the AFO of an hemiplegic elderly with fixed equinavarus deformity who had a femoral neck fracture in his adult life.



Figure 3. Short leg Brace- Ankle foot orthosis - AFO with Klenzak ankle joint, shoe rise and medial T-band for a patient who had leg shortening due to femoral neck fracture and fixed equinovarus deformity.

The avoidance of pressure soars are utmost important. For this purpose the bony prominences that are very sensitive to pressure should be well protected by keeping the walls or the edges of the braces almost 0.5cm away from the bony parts. On the other hand, although, the soft tissues are tolerant to pressure, it must be given evenly and from a wider area to avoid sharp cuts, scrapes or frictions. It should be kept in mind that any mild redness may come up with an open wound in a short period of time due to the decreased circulation and sensation as well as thin and fragile skin of elderly. If the person has diabetes then the care of the skin is much more important and the orthosis should be covered with soft materials for lining such as plastazote to distribute the pressure much more evenly and decrease its effectiveness. However, since hygiene of the orthosis is important, the lining should be cleaned often with a sponge and soap [64].

Corsets

Corsets can be used while the elderly is standing up or walking to support the spine if there is a deformity (kyphosis, scoliosis or both), a fracture or osteoporosis. Although there are different types of corsets (hyperextension corset, Callot corset, Boston, Taylor, Knight or Knight-Taylor braces etc.) they must be comfortable and easily done-and-off (Fig.4). They must not be very tight and must be appropriate to the body of the elderly to avoid pressure sores. Especially, the pressure sensitive areas should be checked carefully while standing up (around costae, pelvis, iliac crista, spina iliac anterior superior) and sitting (around costae, axilla, inguinal area) positions. Besides, the distance between the bony prominences, the edges of the corsets should be rounded and curved for safety.



Figure 4. Hyperextension corset for a patient chronic rheumatoid arthritis having a compression fracture at L1 due to a fall at home.

Shoes

Shoes for elderly are important to provide a better balance and to avoid falls while standing up, standing or walking. They must be lightweight and soft with a flexible sole. The upper part should be wide enough for the toes, preferably without a seam for not to apply pressure on toes. The arches of the feet can be supported according to the need of the elderly with a plastazote or felt insole to distribute the weight evenly to the soles of the feet. The heels can be beveled to increase the support area and their height can be 1.5cm for male and 2-3.5cm for female rather than a flat shoe type to avoid backward leaning of the body and keep the gravity line slightly in front of the knee joint axis on both sides. The slippery sole of the dressed shoes may be one of the reasons of falls especially on wet surfaces.

For this purpose, Oxford-type shoes designed with a rubber tread sole and a treaded bevel heel can be preferred as slip-resistance footwear over dry and wet surfaces. The tread grooves with a depth of 1.2cm are suggested to be perpendicular to the walking direction for a safe walk with the highest coefficients of friction. However, this may not be preferred for the patients with Parkinson's disease who are having tendency to walk with increased friction between the ground and the sole of the shoe that the treads may increase this friction and may force them to fall [65].

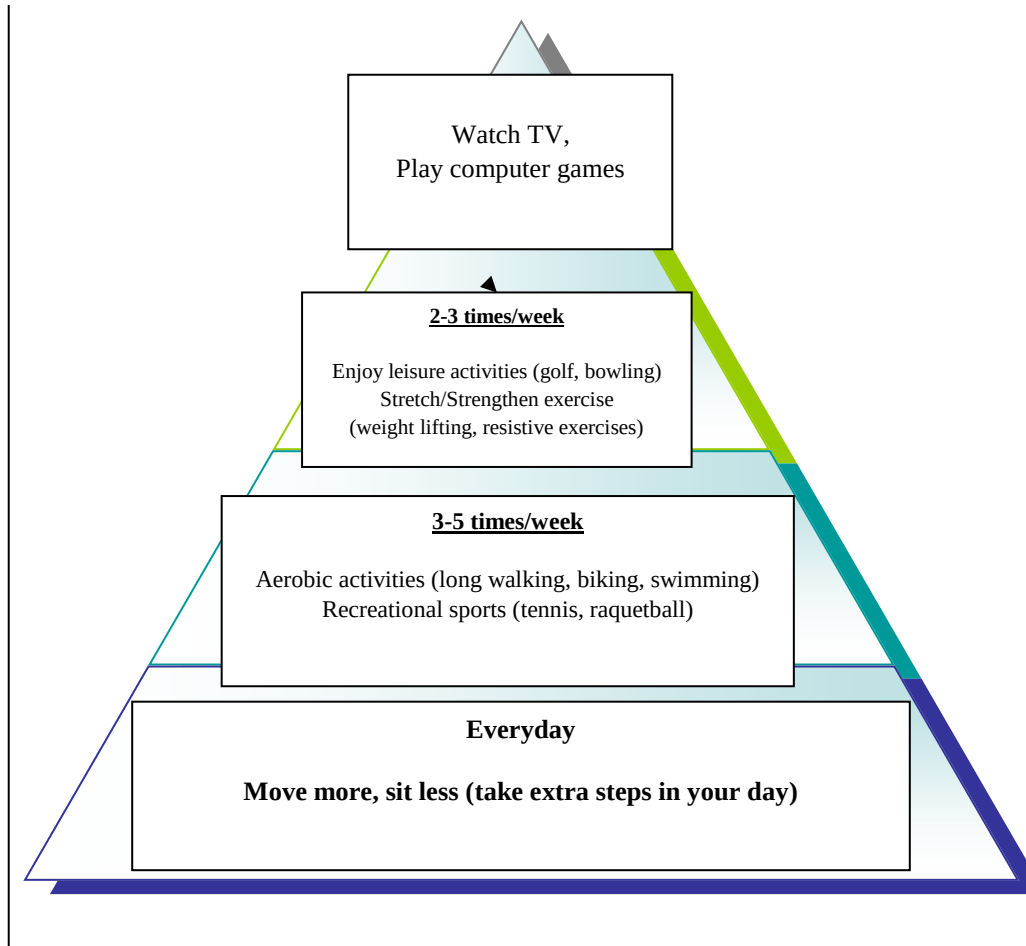


Figure 5. Activity Pyramid (Adapted American College of Sports Medicine. ACSM's Guidelines for Exercise Testing and Prescription, 2006).

Prostheses

An elderly amputee whether is a new or old amputee, should have a well-fitted prosthesis to avoid the undesired complications of pressures that may be given by the socket to the stump. This can be achieved with a well-aligned prosthesis according to the technical rules as well as, to the walking and standing postures of the elderly. On the other hand, the postural changes of an amputee due to aging may require some modifications in the prosthetic fitting alignments for the comfort of the person and also for the better performance in the gait cycle. For example, as the amputee person is getting older the increased anterior shift of the gravity line may result with increased kyphosis and flexion position of the lower limb joints in relation to the weakened antigravity muscles. Although the body may adapt to these angular changes occurring in years, the re-alignment of the prosthesis may be necessary. If the amputee is using an endoskelatal (modular pylon) prosthesis this can be done easily, but if it is a exhoskelatal (conventional) prosthesis then need to be renewed with new alignments.

Since it is much lighter, modular pylon prosthesis can be suggested to the elderly. However, some may not prefer to have changes in their life, continue to use their good old conventional prosthetic limbs. During the decision making on the type of the new prosthesis for the old amputees, their expectations and comfort should be considered widely in relation to their quality of life [66].

Table 7. The characteristics of the lower limb prosthesis according to the level of amputation in elderly

Amputation Level/Type	Prosthesis Type	Socket	Suspension	Prosthetic Foot	Knee Joint	Footwear
Toe amputations	No	No	Shoe itself	No	No	Regular shoe; filling the parts for the toes with plastazote, sponge etc.
Transmetatarsal / Tarsometatarsal-Lizfrank amputation	No	No	Shoe itself	No	No	Light-weight bouts; filling the parts for the toes with plastazote, sponge etc.
Piragof/Boyd Amputation	Below knee prosthesis	Total contact, PTB ¹ / stump end weight-bearing	Bulbous shape of the stump	Piragof/Boyd prosthetic foot or upper part trimmed SACH ² foot	No	Regular shoe ⁴
Syme Amputation	Syme Prosthesis	Total contact, PTB ¹ / stump end weight-bearing	Bulbous shape of the stump	Syme prosthetic foot	No	Regular shoe ⁴
Below Knee Amputation (long and medium stump)	Below knee prosthesis (Moduler Pylon/Conventional)	Total contact, KBM ³	Supracondylar, cuff band and/or medial condylar wedge, if needed,	Hinged ankle joint (5° dorsi flexion - 5° plantar flexion)	No	Regular shoe ⁴
Below Knee Amputation (short and very short stump)	Below knee prosthesis with thigh corset (Moduler Pylon/Conventional)	Total contact, ischial weight-bearing	Supracondylar and cuff and/or Y band	Hinged ankle joint (5° dorsi flexion - 5° plantar flexion)	Side joints of the thigh corset	Regular shoe ⁴
Knee Disarticulation	Knee disarticulation prosthesis (Moduler Pylon/Conventional)	Total contact, ischial weight-bearing + end of stump for new amputees; only end of stump for old amputees	Bulbous shape of the stump	Hinged ankle joint (5° dorsi flexion - 5° plantar flexion)	Side joints	Regular shoe ⁴
Above Knee (long and medium stump) amputation	Above knee prosthesis (Moduler Pylon/Conventional)	Total contact, ischial weight-bearing	Suction and Silasien band	Hinged ankle joint (5° dorsi flexion - 5° plantar flexion)	Single axis locked	Regular shoe ⁴

Table 7. (Continued)

Amputation Level/Type	Prosthesis Type	Socket	Suspension	Prosthetic Foot	Knee Joint	Footwear
Above knee (short and very short stump) amputation Hip dearticulation	Wheelchair	-	-	-	-	-

¹ PTB: Patellar tendon bearing. ² SACH: Solid Ankle Cushion Heel. ³ KBM: Kondular Betting Münster

⁴ Appropriate shoe size and heel height to prosthetic foot.

On the other hand, if it is a new prosthesis for a recently amputated elderly, it should be kept in mind that the lightest and the simplest prosthesis is the best type. Thus, modular pylon prosthesis according to the level of the amputation are the most convenient ones as summarized in table 5.

6. Computer Based Assistive Technology in Geriatric Rehabilitation

Computer aided assistive technology (AT) systems have an extensive role to overcome the ongoing functional and cognitive impairments among elderly [67]. For example, alarming systems to remind the medication time or the action to be done according to the daily living program of the elderly such as getting up and having a short walk in the home or brushing the teeth. Alarming and telemetric information systems, blue tooth signals about the positional and balance changes of the elderly are to gather the information about the daily activity levels of the elderly whether they are in resting situation (sitting, lying down or standing) or active (standing, walking, moving around) in the house. Thus, they can be used for the elderly with their physical, sensorial, mental, visual and auditory assistance to improve their independent living.

Smart Home Technology

Smart home technology systems designed for elderly are the recent advances aiming to gather information about the activity level and daily medical condition of the elderly independently living at home. These systems are designed to give health care at home via special sensors and microprocessors sensitive to movement, sound, vibration, temperature changes etc. located within the furniture, appliances and clothing.

The information of the variety of functional activities can be assessed through the smart home systems, for instance, the heart rate and electrocardiographic changes can be carried out with assessments are done through the tab in the bathroom when the elderly touches to the tab to wash her/his face in every morning. The body temperature can be taken daily through the bed sheet and the body weight through the toilet when they sit on the closet. The amount of daily water taken can be measured via the toilets as the frequency of urinate and weight of the excretion. The mobilization of the elderly can be measured according to the frequency of opening-closing the refrigerator; and the power of turning the tab on/off in the kitchen may

give information about the strength of the upper extremity [68]. Although these facilities seem to be effective in monitoring the elderly in their home while they are living independently, the economical features of setting these systems in the houses are still an important issue for the home care of elderly in many countries. Johnson et al (2007) reported that elderly who may have benefit of smart home settings are not so eager to have these facilities. Thus the motivation and participation of the elderly and the family members to use the technological appliances are important issues of the rehabilitation program. However, simple and uncostly telemetric systems are also available that can be used in any setting, in home or community to follow the health status (pulse, blood pressure, temperature, ECG, etc.), position and mobility status (location, falls, walking etc.) of the elderly. Monitoring the location of the elderly with cognitive impairments is also useful for safety reasons [69].

The care of these sophisticated appliances and their proper usage are also essential interventions. The elderly, their family and the caregivers should be trained properly; an effective orientation period should be given to achieve the overall understanding and the acceptance of the appliances. The control of the system and the control trials of the elderly and the family members are important for the safety precautions.

Conclusion

The geriatric rehabilitation deals with a variety of different conditions effecting the body physically, sensorial, mentally, socially that are overall to increase the quality of life of elderly. The main purpose of this approach is to rehabilitate the emerging conditions due to aging by improving function, balance, coordination; decreasing pain and possibility of falls. Additionally, maintain the existing potential of the elderly having an active life style and provide the utmost precautions to achieve a safety in home or in institutional environments.

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